

Studies in Science Education

ISSN: 0305-7267 (Print) 1940-8412 (Online) Journal homepage: https://www.tandfonline.com/loi/rsse20

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Christoph Kulgemeyer

To cite this article: Christoph Kulgemeyer (2019): Towards a framework for effective instructional explanations in science teaching, Studies in Science Education, DOI: 10.1080/03057267.2018.1598054

To link to this article: https://doi.org/10.1080/03057267.2018.1598054

Published online: 15 Apr 2019.

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STUDIES IN SCIENCE EDUCATION
https://doi.org/10.1080/03057267.2018.1598054

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Towards a framework for effective instructional explanations in science teaching

Christoph Kulgemeyer

Institute of Science Education, University of Bremen, Bremen, Germany

ABSTRACT

Instructional explanations have sometimes been described as an ineffective way to teach science, representing a transmissive view of learning. However, science teachers frequently provide instructional explanations, and students also offer them in cooperative learning. Contrary to the transmissive view regarding explanation, studies suggest that instructional explanations might be successful if they are based on an interaction between explainers and explainees, including the diagnosis of understanding and adaptation to the explainee's needs. The present article has three goals: (1) It will propose a framework for potentially effective instructional explanations, presenting five core ideas of what constitutes effective instructional explanations and two concerning how they should be implemented into science teaching. (2) To justify the framework, the article will review studies on the effectiveness of instructional explanations. It will identify factors that have been researched for their impact on the effectiveness of instructional explanations and discuss them for their applicability to science teaching. (3) This article will connect the research on instructional explanations with the idea of basic dimensions of instructional quality in science. It will discuss the core ideas as particular expressions of the basic dimensions of instructional quality, specifically 'cognitive activation' and 'constructive support'.

KEYWORDS

Teacher explanation; instructional explanation; teaching effectiveness; direct instruction

Introduction

Teachers providing explanations with the intention to 'engender comprehension' (Gage, 1968) are surely a common sight in a science classroom. It has often been argued that explaining is…

(1) …an often conducted practice in all kinds of teaching (Berthold, 2012; Hiebert et al., 2003).
(2) …an important teaching technique in science (Geelan, 2012), for example, because incomplete instructional explanations cause severe difficulties in learning (Weiss & Parsley, 2004).

CONTACT Christoph Kulgemeyer Kulgemeyer@physik.uni-bremen.de
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(3) …a challenging practice, especially for prospective teachers (Charambolous, Hill, & Ball, 2011; Merzyn, 2005). Furthermore, instructional explanations often fail (Wittwer & Renkl, 2008).

Surprisingly, explanation skills are neither the focus of science teacher education nor the focus of empirical research on science education (Geelan, 2012). The primary goal of the present paper is to develop a framework for effective instructional explanations with a focus on science teaching. The purposes of this framework are to give teachers advice on when and how to provide instructional explanations in science teaching and to describe desiderata for further empirical research in science education. The present paper analyses research on verbal explanations because instructional explanations focus on teacher-student communication in the science classroom. Moreover, the focus is on *teacher* explanations, even though the results of the studies might apply to other science communicators and students, as well. Core questions will include: Are there circumstances in which teachers should provide explanations in science teaching? How should they construct these explanations? Previously, instructional explanations have sometimes been described as representing an ineffective, transmissive view of learning (e.g. Chi, Siler, Jeong, Yamauchi, & Hausmann, 2001), prompting the question: Is there such a thing as an effective instructional explanation at all? To answer these inquiries, research in not only science education but also psychology will be analysed. There are many more studies about the effectiveness of instructional explanations in psychology than in science education research. To develop a framework for effective explanation in science teaching, however, these studies must be applied to science-specific prerequisites. It is particularly challenging to interpret these studies regarding the specific challenges of teaching science.

Another purpose of this article is to connect the research on instructional explanations to a concept from the general research on quality of instruction that has been very popular in recent years, the idea of 'basic dimensions' (e.g. Klieme, Schümer, & Knoll, 2001). Many studies have been conducted to find general features of successful instruction. We understand explanation to be but one teaching technique out of many. Therefore, the goal of connecting the features that illustrate the effectiveness of this teaching method to general characteristics of the quality of instruction seems promising. It will likely result in a better connection between two largely independent fields of research.

The present article starts with a discussion of the notion of instructional explanations. Afterwards, we will present the main result of the article, the framework for effective instructional explanations in science teaching. The framework is based on seven core ideas of effective instructional explanations that reflect the current state of research on how these explanations should be provided in science teaching. Of course, this framework requires a detailed literature review for its justification. Therefore, we describe how we built the database of studies used in the literature review. Then we review empirical studies on features of effective instructional explanations and discuss their applicability to science teaching. The core ideas of effective instructional explanation are inferred from these studies and subsequently connected to the basic dimensions of instructional quality. We finish with a discussion on the restrictions of the analysis and implications for further research.

What are instructional explanations? a clarification of the concept

It is important to stress that in science education literature, the term 'explanation' has been used in two different contexts. The first refers to explaining in the sense of a practice used to engender comprehension (Gage, 1968). In this regard, explaining is a communicative, interactive exercise with the goal of helping someone understand science subject matter. Researchers have noted that 'Explanations that are given in educational contexts qualify as instructional explanations because they are deliberately designed for the purpose of teaching' (Wittwer & Renkl, 2008, p. 50). It is important to highlight this 'purpose of teaching' because it differentiates the concept of instructional explanations (or science teaching explanations) from the second important context in which the term 'explanation' is used.

This second context refers to the research on the nature and philosophy of science. In this respect, explanation is viewed as a practice (or even a purpose) of science or scientists that systematically addresses the questions of 'how' and 'why'. Even though many studies have been conducted in this field, there is no generally preferred style of scientific explanation. However, some main characteristics are often agreed upon. For example, 'Many philosophers of science broadly conceptualise scientific explanations as attempts to move beyond descriptions of observable natural phenomena into theoretical accounts of how phenomena unfold the way they do' (Braaten & Windschilt, 2011, p. 641). Even though it is a special example, a scientific explanation in science teaching often means a causal explanation, which involves description of a causal relationship between a phenomenon and an underlying principle. This underlying principle predicts the occurrence of a phenomenon based on a logical connection. The most famous – and still likely the most influential – model of causal explanations is the covering-law model of deductive-nomological explanations by Hempel and Oppenheim (1948). Explanations, in the context of this model, mean a logical deduction of a phenomenon (the *explanandum*) from a general law (the *explanans*) regarding antecedent conditions. The model highlights the importance of logic and deduction in scientific explanations. However, there are many more types of explanations that have been described in the literature. For example, according to Ruben (1990), two other forms of scientific explanations that are frequently used are inductive and deductive statistical explanations (e.g. Rescher, 1962). Deductive scientific explanations use deductive reasoning just like deductive-nomological explanations, but their laws are probabilistic. For example, quantum mechanics is an area that relies on probabilistic laws. Inductive probabilistic explanations derive generalisations from empirical data but lack a deductive cause-effect relationship between a generalisation and the occurrence of a phenomenon (Treagust & Harrison, 1999). Teleological explanations elucidate an observed phenomenon based on its goal or purpose (e.g. Talanquer, 2007). They are frequently used in some scientific disciplines, as well (especially in biology (Yip, 2009)), and there is even evidence that students prefer teleological explanations to causal ones (Trommler, Gresch, & Hamann, 2018). Another very influential model of scientific explanation developed most prominently by Kitcher (1981) understands an explanation in the sciences primarily as an attempt to provide unification for different phenomena. In fact, the structure of explanations has been a topic in science education research for several years, often with the aim of comparing explanation and argumentation (e.g.

Berland & McNeill, 2012; Berland & Reiser, 2008; Brigandt, 2016; Dagher & Cossman, 1992; Osborne & Patterson, 2011; Talanquer, 2007).

Braaten and Windschitl (2011) discuss different concepts of scientific explanations and their importance for science education in more detail. However, that is not the focus of the present article. This article distinguishes merely instructional explanations from scientific ones, and this has consequences. For example, 'The confusion created by the use of the term "explanation" as a synonym for clarification or for communication of reasoning is not merely one of semantics' (Braaten & Windschitl, 2011, p. 655). We agree with Braaten and Windschitl (2011) that scientific and instructional explanations have many differences. Braaten and Windschitl (2011) focus on scientific explanations and do not address the communicative nature of instructional ones. The present paper, however, highlights this communicative nature. Studies describe many epistemological differences between scientific and instructional explanations in science teaching, sometimes simply referred to as 'science teaching explanations' (Martin, 1970; Treagust & Harrison, 1999). Geelan (2012) summarises the differences as follows:

> 'Science explanations are strictly characterised as theory and evidence-driven, use correct scientific terminology[,] and can include analogical models.
>
> Science teaching explanations differ in rigour, length[,] and detail, involve varying degrees of "explain how" and "explain why", are sometimes open-ended, include human agency[,] and can raise new questions as they answer previous questions.' (p. 988)

Briefly put, an instructional explanation has the specific goal of communicating scientific content. A scientific explanation has a different intention, which is, according to Braaten and Windschitl (2011), providing a theoretical account of how and why phenomena occur, e.g. by presenting a logical connection between *explanans* and *explanandum*. Perhaps a logical connection helps an individual to understand, but this is not self-evident. This paper investigates instructional explanations and therefore focuses on teacher-student communication in science teaching. What makes a verbal explanation likely to fail or be understood?

The common assumption that an instructional explanation is nothing more than a concise and precise presentation of a scientific explanation is seductive; knowledge regarding the structure of science helps considerably with constructing these kinds of explanations, and no pedagogical content knowledge is required. This assumption is deceptive, however, and there is evidence to support this claim. Empirical research shows that putting the structure of science first when explaining to a novice leads to less understanding than using the learner's prerequisites as a guideline (Nathan & Petrosino, 2003). Sometimes, content experts are not aware of that and are – despite their high content knowledge – less successful at explaining than people with lower content knowledge. This is sometimes referred to as an 'expert blindspot' (Nathan & Koedinger, 2000). It is a common misunderstanding that instructional explanations equal scientific ones that are presented clearly, even in research. For example, Chi et al. (2001) use the term *explanation* with a negative connotation (i.e. as an obstacle to achievement) because students are not actively involved. One might even assume that this leads to the described gap between the importance of the practice and the low number of related studies in science education. Explanations have the reputation of being less

effective than student-centred teaching techniques because merely presenting a scientific explanation actually *is* less effective in terms of reaching an understanding of the subject matter (Wittwer & Renkl, 2008). Effective explanation, however, is much more than a presentation. That does not mean that using a clear structure to construct explanations leads to low achievement. A well-structured instructional explanation can be beneficial (see the section entitled 'Structure of the explanation: Rule-example or example-rule'). However, the structure of the science is not everything if the primary focus should be on a communication process. A communicative view of explanation always starts with an analysis of both the scientific content that needs to be explained and the prerequisites of the explainee, e.g. his or her prior knowledge (Kulgemeyer & Schecker, 2009).

This is very challenging, especially for novice teachers. Merzyn (2005) reports on an interview study with instructors who had just finished an academic teacher education programme. He notes that they perceive explanation as the most challenging part of their new duties. This suggests that learning how to explain is indeed very rarely part of teacher education in the sciences, even though there are examples in mathematics teacher education (Charalambous, Hill, & Ball, 2011). We can only speculate about the reasons, but one might be that teacher education focuses on student learning and methods of activating students, while teachers' explanations sound like they are part of a strong didactical tradition of instruction. To this claim, Geelan (2012, p. 987) simply answers that 'teacher explanations are not necessarily antithetical to inquiry learning or tied to lecturing'.

Interestingly, students highly value teachers with advanced explanation skills. Wilson and Mant (2011a) report on a study using a questionnaire administered to 5,044 year 8 students of 153 science teachers in England. Their goal was to determine the characteristics of the teachers who had been praised the most by their students (i.e. 'exemplary teachers'). They report that the greatest difference between exemplary teachers and average ones lies in the agreement of their students with the statement that 'my teacher explains scientific ideas clearly'. Wilson and Mant (2011a) summarise that explanation is a core skill of exemplary science teachers. In a subsequent study, Wilson and Mant (2011b) invited six of the eleven exemplary teachers to a group discussion to explore their perspective on what skills an outstanding teacher possessed. Surprisingly, even though these teachers were praised for their explanation skills, they did not mention explaining as a key factor of the quality of their teaching.

This perhaps corroborates the assumption that teacher explanations have the reputation of being old-fashioned, ineffective ways of teaching, although they sometimes help engage students in the sciences (see the literature review below). It may also indicate that explaining is widely misunderstood as the behaviouristic, transmissive practice of *presenting* scientific explanations to students. Effective explanation is a constructivist practice that involves adaptation to the explainees' needs and interaction between the explainer and explainees (Wittwer & Renkl, 2008).

In the following, we will report on studies that understand explanation as a constructivist practice and discuss their effectiveness in detail. Prior to the literature analysis, we want to present a framework for effective instructional explanations in science teaching that is subsequently supported by said analysis.

A framework for effective instructional explanations in science teaching

Based on the studies we analyse below we propose core ideas of what makes instructional explanations effective. Furthermore, we propose a model of the process of effective explanation in science teaching that describes the most promising way to implement instructional explanations based on the state of the literature (Figure 1). Of course, future studies might lead to refinement.

These core ideas are the basis of the framework. We understand them as the underlying concepts of the empirical studies we report on in the following sections. The seven core ideas are:

1. It is important to adapt the explanation to the prior knowledge and the misconceptions of the explainees (e.g. Wittwer, Nückles, Landmann, & Renkl, 2010). It is likely also important to adapt the explanation to their interests, even though the empirical evidence is weaker for this claim than for the first (Kulgemeyer & Schecker, 2009).
2. To adapt the explanations, the field of science communication provides four promising variables: language level, examples/analogies, level of mathematisation, and representation forms and demonstrations (Kulgemeyer & Schecker, 2009, 2013).

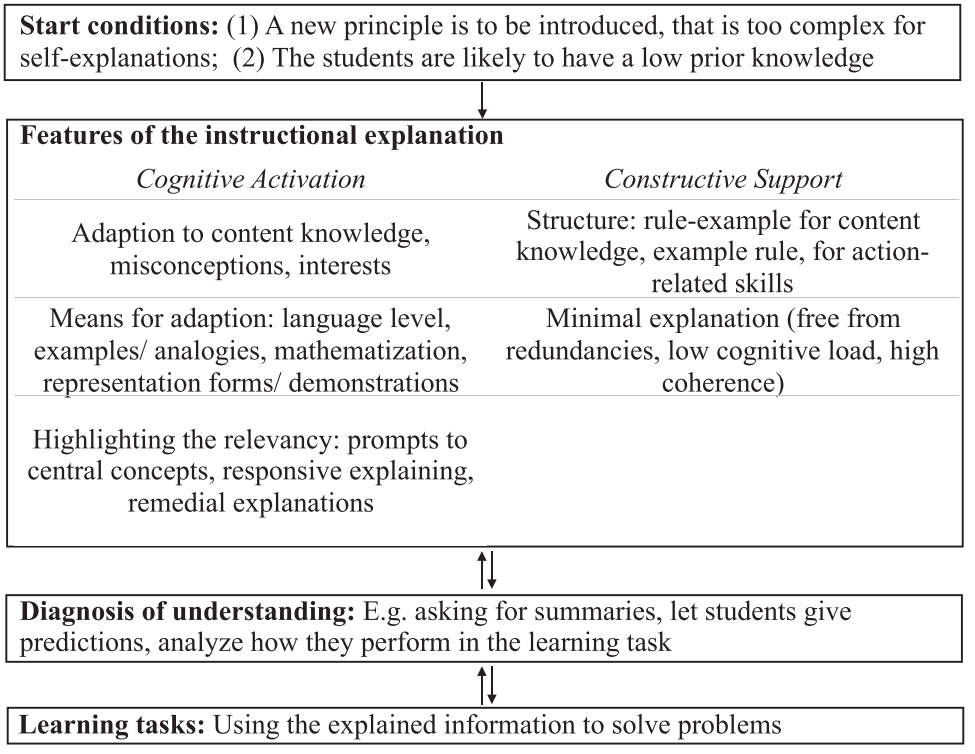


Figure 1. Framework of effective instructional explanations in science teaching. An instructional explanation begins with checking starting conditions and continues with providing a first explanation. The first explanation is followed by a diagnosis of understanding, and finally, a learning task.

3. It is important to highlight the relevance of the explained content to engage the explainees in deeper processing of the information. Possible ways to highlight its relevancy are prompts to central concepts (Berthold, Röder, Knörzer, Kessler, & Renkl, 2011; Chi et al., 2001; King, 1999), responsive explaining (e.g. Diakidoy, Kenedeou, & Ioannides, 2003), and remedial explanations that explicitly note the addressed misunderstanding (Roelle, Berthold, & Renkl, 2014; Sánchez, García-Rodico, & Acuña, 2009).
4. The structure of the explanation impacts its effectiveness. This depends, however, on the learning intention (Seidel, Blomberg, & Renkl, 2013). A 'rule-example structure' is likely more appropriate for gaining content knowledge (Tomlinson & Hunt, 1971). An 'example-rule structure' is presumably more appropriate for developing action-related skills such as experimenting or using a procedure to solve a specific type of science-related problem (Champagne, Klopfer, & Gunstone, 1982).
5. Explanations should be focused on what matters and avoid digressions (e.g. 'minimal explanations') (Black, Carroll, & McGuigan, 1987). They should be free from redundancies because these are suggestive of low relevance (Anderson, Corbett, Koedinger, & Pelletier, 1995). They should also require a low cognitive load; however, whether, for instance, an additional example is perceived as cognitive load or as a useful illustration depends on the prior knowledge of the explainees. Explanations should also have a high coherence that helps connect their elements and build a mental image (Wittwer & Ihme, 2014).

The most interesting question for science teachers is surely when and how to provide instructional explanations. Empirical research provides some guidelines to answer that inquiry, as well:

6. Instructional explanations should only be given if self-explanations are likely to fail (Renkl, 2002). This is the case if a complex principle is to be explained that is new to students. Particularly in science, it is also the case if students are likely to hold many misconceptions regarding the principle in question, because an instructional explanation might help avoid an illusion of understanding as opposed to self-explanations (Chi, Bassok, Lewis, Reimann, & Glaser, 1989; Chi, de Leeuw, Chiu, & LaVancher, 1994). Students sometimes stop self-explaining activities when they *think* they understand the topic, but more often than not they objectively do not understand it.
7. An instructional explanation is not finished after the presentation of information. It should be followed by a diagnosis of understanding that allows further adaptation (e.g. by asking students for summaries or analysing their performance in a learning task). After the diagnosis and a possible remedial explanation, students necessarily need time to use the explained information to solve related problems on their own. This phase is essential for an instructional explanation to be successful (Altmann & Nückles, 2017; Webb, Ing, Kersting, & Nemer, 2006; Webb, Troper, & Fall, 1995).

These seven core ideas result in a framework of effective explanation in science teaching (Figure 1).

The framework describes explanation as a process that consists of four steps:

First step: According to the framework, first, teachers should be aware of learning intentions. Instructional explanations are only effective if they focus on principles, not on step-by-step descriptions of how to solve problems. Therefore, a starting condition for instructional explanations should be a principle that needs to be introduced. This principle should be too complicated to allow self-explanations, meaning that self-explanations on this topic would most likely result in an 'illusion of understanding'. This is likely the case for principles in science in which individuals have many misconceptions. Newton's third law is a good example of such a principle. A second starting condition is that the learners should have low prior knowledge of the explained topic. If the students already possess high knowledge about a subject, the teacher should use strategies that help them construct self-explanations, instead.

Second step: After these starting conditions have been checked, a 'circle of explaining' starts. An initial instructional explanation should be presented that includes the subject's core ideas and fulfils high standards in cognitive activation and constructive support (see the section entitled 'Connections to the idea of basic dimensions in the research on instructional quality').

Third step: This initial explanation is followed by a diagnosis of understanding, which is essential for further adaptation. Promising ways to evaluate comprehension are, for example, asking students for summaries of the explanation, letting them predict the occurrence of a phenomenon (e.g. in a demonstration), or analysing how they perform in a task. After this diagnosis, the teacher can reconstruct the mental model he/she has about the explainee's needs. This could result in a better-adapted explanation or a remedial one. He/she should strongly avoid simple repetition because that certainly would not result in deeper processing of the information, but rather in a high cognitive load. This sequence of better-adapted explanation and diagnosis might occur several times and focus increasingly on details. However, repeating it too often does not seem to be a good idea. Based on the current research, we cannot deduct a rule of thumb regarding how many repetitions are acceptable. This likely depends on both the content and the learners.

Fourth step: After this phase of explanation and diagnosis, a learning task should follow that requires use of the information in question to solve problems. Of course, performance in this task could be another source for further diagnosis and even for better-adapted instructional explanations.

It is no surprise that this model of effective explanation in science teaching shows many similarities to the direct instruction method of education. Hattie's (2009) famous meta-analysis describes direct instruction as a compelling teaching method. It consists of seven steps: (Hattie, 2009, 204–206):

(1) The teacher should be aware of the learning intentions
(2) The teacher should understand the success criteria
(3) The teacher should build commitment and engagement
(4) The teacher presents the information in various ways
(5) The students engage in guided practice and work with the new information under direct supervision

(6) The teacher (assisted by the students) summarises the most important parts of the lesson
(7) Finally, the students have a chance to engage in independent practice

Of course, these seven steps are design principles for a whole lesson, not just for an instructional explanation. However, Steps Four and Five are basically what successful explaining is all about: adapted instructional explanations based on diagnosis and followed by learning tasks. The importance of Step 1 should be stressed again in the context of instructional explanations. Based on Hattie's (2009) analysis, direct instruction is a powerful method if the learning intention is subject matter knowledge. From our perspective, for other learning intentions in science (e.g. communication or experimental skills), direct instruction is likely not the method of choice, and the same should be true for instructional explanations.

In the following sections, we will analyse the literature for empirical evidence that supports or questions this framework.

Method

For the present article, it is a central challenge to include all the relevant publications in the literature review. To ensure that we identified the most important ones, we followed the steps used in the review study by Belova, Rundgren, and Eilks (2015). We started by analysing prior reviews of instructional explanations (Geelan, 2012; Renkl et al., 2006; Wittwer & Renkl, 2008) to ensure we selected the most important keywords for the database search. Wittwer and Renkl (2008) present a very detailed review of psychological studies on the effectiveness of instructional explanations that served as an excellent starting point. Geelan's (2012) paper in the *Second Handbook of Science Education* added science-specific ideas, especially concerning the differentiation of the scientific explanations and science teaching explanations we discussed above. We then started searching both the Web of Science and ERIC to find a database that included all possible contributing studies. To reach this goal, we initially used keywords that were as broad as possible: 'instructional explanation', 'teaching explanation', and 'teacher explanation'. However, we found as many as 8,019 studies following this approach (we stopped including papers in the database on 23 January 2018). In the next step, we looked for duplicates since it was impossible to deeply analyse such a large volume of studies. After eliminating all the duplicates, we decided to focus on peer-reviewed studies only. The goal was to review the empirical evidence of the effectiveness of instructional explanations. We were more likely to find the high-quality studies we wanted to base our analysis on in peer-reviewed journals or books. After this step, we ended up with 1,012 publications.

In the next step, we reduced the sample by searching for the exact combination of 'instructional explanation', 'teaching explanation', or 'teacher explanation'. The reason for this decision was that in this stage, we included many articles that focused on scientific explanations, often on teaching students how to construct them (e.g. McNeil, 2009). Some of these studies dealt with the structure of scientific explanations in textbooks (Velentzas & Halkia, 2018), with students' preferences for certain types of scientific explanations (Trommler et al., 2018), or even learning progressions of scientific explanations (Yao &

Guo, 2017). We decided not to consider these studies for further analysis. The difference between scientific and instructional explanations has been discussed above. This process led to a sample of 175 studies that dealt with instructional explanations.

We analysed the abstracts of these studies and compared the sample with the literature referenced in the frequently cited review paper by Wittwer and Renkl (2008) to ensure that we found the most relevant studies. We also compared the results with the review paper of Geelan (2012), which focuses on teaching explanations in science, and with our own database. The latter led to including a number of German studies.

Since our focus was to analyse the studies on instructional explanations in science teaching, we searched for the combinations 'instructional explanations science', 'teaching explanations science', and all combinations of 'instructional explanations' and 'teacher explanations' on the one hand, and 'physics', 'chemistry', and 'biology' on the other. In doing so, we wanted to ensure that we included important studies from science education in the database, which resulted in the inclusion of, for example, Geelan (2013) and Cabello (2013). After once again erasing the duplicates, this final step led to a database of 156 studies that we included in the following review of the literature to develop the framework of effective instructional explanations in science teaching.

What makes instructional explanations effective? a review of relevant literature

Wittwer and Renkl (2008) wrote the most important review that focuses on identifying what makes instructional explanations effective in general. They describe four essential factors that lead to sufficient instructional explanations:

(1) adaptation of the explanation to the learner's knowledge,
(2) a focus on concepts and principles,
(3) integration into ongoing cognitive activities, and
(4) knowledge-construction should not be replaced by explanations.

Adopting these factors in science teaching seems promising, of course. Since 2008, more research has been conducted, and the literature review reveals more factors that should be added to the list. Furthermore, Wittwer and Renkl (2008) do not refer to science education research. There are studies in science education that need to be considered, such as those discussed in the review by Geelan (2012). Based on the reviews of Wittwer and Renkl (2008) and Geelan (2012), as well as the studies in our database, we identified nine factors that were focused on in prior studies. They have been described as likely to have an impact on the effectiveness of instructional explanations.

Factor 1: the structure of the explanation: rule-example or example-rule?

As we have discussed above, the structure of an explanation is an important issue when distinguishing between scientific and science teaching explanations. For example, one could wonder whether a phenomenon was being explained by an underlying rule, or ask the opposite question: Is an underlying rule being illustrated by an example? For a verbal instructional explanation in science teaching, a teacher must

decide whether he/she wants to derive a principle from an example that is the starting point of an explanation or whether the principle should be presented first. After presentation of the principle, examples might be given that help students understand it. Research exists regarding whether the 'inductive' or 'deductive' strategy is the more effective in instructional explanation. In educational psychology, these two approaches have been researched under the names 'example-rule strategy' and 'rule-example strategy'. They have been studied primarily in the context of the use of video examples for acquiring teaching skills (Korthagen & Kessels, 1999). The evidence points to both strategies having different effects. If *declarative* content knowledge is the goal, it seems more effective to present a rule at first and illustrate it with an example afterwards (Tomlinson & Hunt, 1971). However, for *procedural* knowledge or learning routines, the example-rule strategy turns out to be the better choice (Champagne et al., 1982). In a more recent study, Seidel et al. (2013) compared both strategies in teacher education and arrived at similar results. They randomly assigned a small group of 54 pre-service teachers to one of the two strategies and measured (a) factual knowledge, (b) observation and evaluation of classroom situations, and (c) lesson planning skills before and after the intervention of the two instruction strategies. They found that the rule-example group outperformed the example-rule group in factual knowledge ($t(40) = 5.87$, $p < .05$, Cohen's $d = 1.86$) and in observation and evaluation of classroom situations ($t(47) = 1.89$, $p < .05$, Cohen's $d = .55$). However, in lesson planning, the example-rule group scored significantly higher ($t(46) = -3.08$, $p < .05$, Cohen's $d = .91$).

What do these results mean for instructional explanations in science teaching? Quite often, content knowledge is the goal of an explanation, e.g. the understanding of a specific principle like Newton's first law. In that case, the rule-example strategy seems to be the method of choice. However, if more action-related learning intentions apply, the example-rule strategy would be superior. In science teaching, that might be the case if experimental skills or ways to solve specific scientific problems are learning intentions. However, for experimental skills, instructional explanations are certainly not the first choice. It is important to acknowledge such differential effects – even more so since Geelan (2013) found that high school physics teachers often prefer starting with a short qualitative imagination of a situation and afterwards swiftly shift to mathematical descriptions. That approach seems to be essentially a specific application of the example-rule strategy and therefore is not the method of choice in all cases.

However, we are not aware of any studies on this topic in the domain of science. It might still be a promising strategy to present an example that shows why a principle is important at the beginning of an explanation. This could help motivate students to follow the elucidation. Research on context-oriented learning in science supports this claim (e.g. Fechner, 2009). Still, in the following explanation, the principle does not need to be derived from this example. However, it might be presented anyway and illustrated subsequently by different illustrations. The first example simply serves as a 'hook' in this case.

Factor 2: adaptation to the explainee

It has often been argued that adaptation to a learner's knowledge level is one of the critical strategies for effective explanation. Wittwer and Renkl (2008) state: 'Taken together, there is strong evidence that instructional explanations – either provided in interactive or individual settings – should be adapted to a learner's specific knowledge level' (Wittwer & Renkl, 2008, p. 53). Even an ordinary approach to explanation would support the point of view that the presentation of information should be adapted for learners with low content knowledge. Of course, these learners do not know how to use, for example, scientific diagrams. They also do not know the technical terms in a new area of study. In this case, explaining means using language and concepts with which these learners are familiar. However, research shows that the need for adaptation is not only present regarding explanations given to novices, but also for those provided to experts. Instruction that is as easy as possible is sometimes even a hurdle for learners with high content knowledge, a phenomenon that has been described as the 'expertise reversal effect' (Kalyuga, 2007; Kalyuga, Ayres, Chandler, & Sweller, 2003; Kalyuga & Renkl, 2010). A possible explanation for this effect comes from cognitive load theory (Mayer, Moreno, Boire, & Vagge, 1999; Sweller & Chandler, 1991). While some of the presented information might be new for novices, it could be redundant for experts. This information, therefore, is more of a cognitive load than it is helpful for these experts (Kalyuga, 2007) because it uses cognitive resources that are no longer available for processing the actual content. Additionally, research on text comprehension shows similar results. McNamara and Kintsch (1996) found that texts that are more complex are more useful for learners with high content knowledge than those that are theoretically optimised for comprehensibility. These texts, however, are more useful for novices.

Duffy, Roehler, Meloth, and Vayrus (1986) found that outstanding teachers can assess students' difficulties and use the results to adapt their explanations. Diagnosis seems to be an essential skill for good instructors in terms of being able to diagnose misunderstanding and use this knowledge for adaptation. For example, students providing explanations through cooperative learning techniques such as the jigsaw method learn significantly more in the areas they explain than in the areas their peers explain to them (Berger & Hänze, 2015; Hänze & Berger, 2007). It has been suggested that the reason for this is a missing adaptation (Wittwer & Renkl, 2008). Perhaps the problem starts with missing diagnostic skills: Whether information about prior knowledge increases students' explanation quality is an open question (Nückles, Wittwer, & Renkl, 2005). The importance of diagnostic skills is also stressed by studies of human tutoring. On the one hand, tutees do not uncover their difficulties in understanding usually (Herppich, Wittwer, Nückles, & Renkl, 2014), but on the other hand, inexperienced tutors very rarely assess their tutees' knowledge of their own accord (Chi, Siler, & Jeong, 2004; Wittwer et al., 2010). Additionally, explainers often think that their explainees know more than is actually the case (c.f. Nickerson, 1999 regarding the 'false consensus effect'). Wittwer et al. (2010) gave computer experts information about their tutees' prior knowledge and showed that these experts were able to adapt their instruction accordingly. Moreover, for tutees with high prior knowledge, the computer experts emphasised

connections between concepts, which should enable the tutees to engage in deeper learning processes.

For science teaching, the state of the literature suggests that adaptation to learners' prior knowledge is crucial for instructional explanations. However, it is unclear whether this adaptation is only vital regarding cognitive prerequisites. Whether adaptation regarding learners' interests and motivation is an essential factor, as well – as Kulgemeyer and Schecker (2013) suggested – needs to be further researched. Geelan (2013) even sees humour as a form of adaptation to learners' interests. In his study, he identified humour as a frequent element of explanation.

A promising example for adaptation in science teaching is the use of adapted primary literature (APL) (e.g. Phillips & Norris, 2009) as an approach to bridge the gap between the languages of science and school science (sometimes also referred to as the language of education). This approach understands reading as an element of inquiry and aims to enable students to learn about authentic speech acts in science. APL follows the authentic form of scientific papers, but is written adapted to the leaners (Ford, 2009). The prose used, therefore, combines elements of the languages of science and school science, but emphasises the first.

Factor 3: minimal explanations

A specific form of adaptation that has been discussed in the literature is minimal explanation. 'Minimal' means providing just the information that is needed (and not more), which is highly individualised. Effective explanations have been described as precise, coherent, and free from irrelevant details (Anderson et al., 1995), which epitomises the idea of minimal explanations. Renkl et al. (2006) note that this criterion comes from research on computer-based learning (e.g. 'minimal instruction', Carroll (1990)), where it has been shown that reducing texts increases their comprehensibility (Black et al., 1987). Additionally, redundancies between information from different sources should be avoided (Mayer & Moreno, 2003). The reason for this, again, can be seen in the cognitive load theory: Redundancies require cognitive resources, making them unavailable for deep-processing.

Another critical factor of minimal explanations is their coherence. Constructing a mental representation of the explained content requires building connections between its elements (e.g. propositions) (Kamalski, Sanders, & Lenz, 2008). 'Coherence' essentially means a high density of these mental connections. It has sometimes been distinguished from 'cohesion', which is a characteristic of a text that supports a coherent mental image. García-Sanchez and Fidalgo-Redondo (2006) present a helpful overview of cohesion indicators from the perspective of linguistics (Table 3, p. 186). A simple example of high cohesion would be the occurrence of the same noun in two sentences, which helps to connect these phrases (in this case, as a lexical tie). Another way to increase the cohesion of a text is the use of connectives, such as 'before' (which suggests a temporal connection). For explanations, the connective 'because' is the most important because it links *explanans* and *explanandum*. 'Because' is a causal connective, and these are sometimes assumed to be the most relevant (Lehman & Schraw, 2002) and plausible form of relation (Connell & Keane, 2004). Cohesion has often been discussed as important for written explanations (e.g. Wittwer & Ihme, 2014), but there is no reason that it should not play an essential role in verbal explanations, as well.

Factor 4: remedial explanations

Adaptation also plays a central role in so-called 'remedial explanations'. While instructional explanations often focus on new content or provide first insights into a new topic, remedial explanations occur when a teacher focuses on misunderstandings or mistakes (Sánchez, García-Rodico, & Acuña, 2009). Combining instructional and remedial explanations in such a way that the new topic is being introduced through instructional explanations has been proposed. A diagnosis of faulty knowledge would follow, and afterwards a remedial explanation would be given that was adapted to this problem (Roelle et al., 2014). The effectiveness of remedial explanations depends on the adaptation used (Leinhardt & Steele, 2005). It is also contingent on the teachers explicitly highlighting which specific misunderstandings they want to focus on in their remedial explanation. This emphasises the perceived relevancy of the remedial explanation (Acuña, García-Rodicio, & Sánchez, 2011). The perceived relevancy, however, seems to be the crucial prerequisite. Whether the remedial explanation is perceived as relevant influences students in their decision to follow the explanation or ignore the communicative attempt (Acuña et al., 2011).

Factor 5: follow-up learning tasks

There is an ongoing discussion about the effectiveness of instructional explanations compared to self-explanations. For example, in research regarding tutoring, it has often been stressed that self-explanations are more effective than instructional ones because of the possibility of engaging learners in the more active processing of information (Chi, 1996). Sometimes, the role of instructional explanations has been described as merely supporting self-explaining activities (Renkl, 2002). However, it has been shown that self-explanations have a higher probability in resulting in a so-called 'illusion of understanding' (Chi et al., 1989, 1994). Students tend to stop their self-explaining activities at a point where they think they understand the topic, even though they objectively do not comprehend it. During instructional explanations, teachers have better control in avoiding an illusion of understanding (Renkl et al., 2006). This problem seems especially evident in science. Research on students' misconceptions has revealed many significant misunderstandings of various topics in the sciences that are often based on personal experience (c.f. the bibliography of Duit & Pfund (2009), which includes more than 8,400 entries). It is difficult to change these misconceptions, if doing so is possible at all (Duit & Treagust, 2003; Posner, Strike, Hewson, & Gertzog, 1982). Self-explanations of topics with many misconceptions might, therefore, often result in an illusion of understanding or even in a more firmly established misinterpretation.

To bridge the gap between active processing and instructional explanations, integrating the latter into cognitive activities has been suggested (Wittwer & Renkl, 2008). Some tools have been discussed that might result in a deeper processing (see below, the section entitled 'Tools to increase the quality of presentation'). Adaptation to the learners' interests is a promising strategy, as well. The way that instructional explanations are embedded in teaching seems to be important, also; learning tasks that use the explained knowledge for problem-solving activities contribute considerably to the effectiveness of the explanation (Altmann & Nückles, 2017). In this sense, follow-up learning

tasks can be considered an essential component of an explanation. An instructional explanation is more than just a presentation of knowledge, but rather a well-designed mixture of an explanation in the narrower sense (an initial presentation of information), which is combined with a diagnosis that results in adaptation and follow-up tasks, which then require the presented knowledge to solve problems.

Various studies have been conducted to examine the effects of different ways of engaging students in cognitive activities after an initial presentation of information (Webb & Paliscar, 1996). It has been shown that follow-up activity was even more important for predicting learning outcomes than the explanation in the narrower sense itself (Webb & Farivar, 1999; Webb et al., 2006). Moreover, Webb (1992) and Webb et al. (1995) corroborate this claim by showing that learners who engage in problem-solving activities after an explanation reach higher learning outcomes than those who only listen to it. Renkl et al. (2006), therefore, consider self-guided, further elaboration of explained information to be crucial in integrating presented information into one's knowledge base. They also note that instructional explanations are often a reaction to students' questions (e.g. they are offered if students realise a problem and ask for clarification). However, even very well-crafted explanations are ineffective if they refer to questions learners forgot during listening to them. Follow-up learning tasks are also a possible tool to help learners remember the reasons they asked for an explanation in the first place.

Follow-up learning tasks should be designed in a way that allows learners to actively engage with information. Chi (2009) proposed a framework that differentiates between the ways learners deal with instructional materials. Active tasks for learners would, for example, include highlighting or summarising important parts of an explanation on their own. Regarding science teaching, the literature on problem-based learning also provides insights into how to design these tasks (for an overview, see Hmelo-Silver (2004)).

Factor 6: prompts to focus central concepts

One strategy that has been proposed to engage learners more actively with the explained content is prompting. Prompts are questions or hints that are thought to lead to deeper processing (Chi, 2000). Berghmans, Michiels, Salmon, Dochy, and Struyven (2014) stress the importance of prompts by differentiating between a facilitative style (which includes prompting) and a merely directive approach to teaching. The facilitative style also provides explainers with the information they need to adapt their explanations. Chi et al. (2001) present many examples of general prompts and propose trainings in which tutors (or explainers) can use them. Berthold et al. (2011) found that giving instructional explanations in combination with prompts correlates with a higher knowledge gain.

An important contribution to the research on prompts in instructional explanations comes from Roelle et al. (2014). They compared two strategies that have often been proposed for instructional explanations: providing remedial explanations that adapt to misunderstandings on the one hand, and providing prompts that focus on the deeper processing of central concepts on the other. They report a large statistical effect of prompts on learning outcomes ($F(1, 76) = 14.07$, $p < .001$), but, surprisingly, no effect regarding remedial explanations ($F(1, 76) = 0.87$, $p = .353$). They further analyse learning

processes and conclude that the prompts 'elicited inferences on the central content of the explanations' (Roelle et al., 2014, p. 223) and these inferences cause the higher knowledge gain. The authors present different interpretations concerning the missing effect of remedial explanations on knowledge gain. One is that learners do not see the need to engage in deep-processing. This would also support the claim discussed above that remedial explanations are only effective if they are perceived as relevant (e.g. due to teachers explicitly noting which type of misunderstanding they address).

Relevance indeed is an important point: Remedial explanations might be perceived as repetitions of the initial explanation if it is unclear that they provide something new. In this case, they might simply lead to a higher burden, since redundancies increase learners' cognitive loads and do not lead to deeper processing of the explained information. This supports the assumption that it is an ineffective strategy for a teacher to start with a minimal explanation, diagnose problems afterwards, and then present remedial instruction. This might appear redundant. Teachers should at least name the diagnosed problem and note that the subsequent explanation is designed to deal with it. We further elaborate on this in the section entitled 'Connections to the research on instructional quality' below.

Also, Acuña, Garcia-Rodicio, & Sánchez (2011) emphasise the relevancy of highlighting misunderstandings explicitly before explaining (even though they talk about instructional explanations in general, and not remedial explanations in particular). They explain the effect using the 'mental model repair view' (Chi, 2000). According to this theory, learning requires the detection of gaps in the knowledge and generating explanations to repair them. Noting the diagnosed problem after an initial instructional explanation might help in detecting gaps, and remedial explanation might provide the knowledge to repair them.

Factor 7: responsive explaining: highlighting the relevancy

Similar to the discussion on remedial explanations, another aspect of instruction has been discussed in recent studies. 'Responsive-explaining' might contribute to effective explanation. This term refers to explanations that explicitly indicate that a learner needs them. Several studies support the claim that responsive explanations are a way to provide effective clarification. Sánchez et al. (2009) report on a study in which they compared rough explanations of a geology topic with the same explanations that were supported by prompts. Their research makes the reason this type of explanation is important obvious: It helps learners overcome widespread misconceptions. Notably, the learners who received explicit prompts outperformed the control group in the pre-test.

This study seems to provide substantial evidence for science education, specifically. The research on misconceptions, as discussed above, is of great importance for science teaching. If highlighting misconceptions improved the quality of explanation, it might be an important strategy for instructional explanations in science. The basic idea is also very close to conceptual change theory, in which noting misconceptions and later restructuring them is the core of many strategies that aim to achieve conceptual change (c.f. Scott, Asoko, and Driver (1992)). There seems to be a connection between the theory of conceptual change and the theory of instructional explanations. Some studies on

responsive explaining also focus on scientific content. Diakidoy, Kenedeou, and Ioannides (2003) provided written explanations about the concept of energy. For one group of learners, the explanation was enriched with an explicit hint regarding a common misconception concerning energy: that many learners understand energy as something material. The other group did not receive this hint and were outperformed by the experimental group. Kendeou and van Den Broek (2007) focussed on Newton's laws and used the same strategy as Diakidoy, Kenedeou, and Ioannides (2003). In their study, those learners receiving the hint outperformed the control group, as well. These hints possibly emphasised the relevancy of the explanations and led to deeper processing of the information. As there has been considerable research on misconceptions in science education, giving these hints might be rather easy for science teachers.

Acuña et al. (2011) report on an interesting finding concerning responsive explaining. They compared the effect of responsive explanations on learners with different levels of prior knowledge. In their experiment, they referred to a small sample of learners with (a) low content knowledge ($N = 39$) and (b) high content knowledge ($N = 38$). They found a meaningful and significant impact of responsive explanations in the group of learners with low content knowledge; however, those with high content knowledge did not benefit. The authors interpret their findings according to the mental model repair view mentioned above. Learners with low content knowledge need explicit hints on what aspects of their prior knowledge are intended to be revised by an explanation to engage in deeper processing. Learners with high content knowledge, however, do not need that information and therefore cannot benefit from it. Wittwer and Renkl (2008) mention that explanations for learners with high content knowledge are more likely to fail in any case. Combined, this seems to support the assumption that instructional explanations should be given only to learners with low content knowledge. In contrast, learners with high content knowledge benefit more from self-explaining (Wittwer & Renkl, 2008). Science teachers, therefore, might consider giving instructional explanations when they introduce a new concept, but not at a later stage of a teaching unit.

Factor 8: focusing on concepts or principles

Renkl et al. (2006) propose the claim that instructional explanations should only be given if they focus on principles or functions and note that in science teaching, principles are more relevant. They argue that understanding principles helps learners build cognitive schemes that can be used to solve different problems by applying the same concept. Indeed, this seems to be important for science teaching since it comes close to the essence of the nature of science. Many form of scientific explanation – as discussed above and not necessarily a science teaching explanation – usually explain a phenomenon by using a principle that helps predict its occurrence. Finding principles that explain the occurrence of different phenomena is crucial for science itself and for scientific explanations, specifically (c.f. Kitcher (1989) and the unification theory of explanation).

However, research shows that explainers sometimes tend to clarify the underlying principle less and instead provide more instruction regarding how to solve a specific problem without indicating why (Alty & Coombs, 1981). In the computer sciences domain, Dutke and Reimer (2000) found that including the relevant principles in an

explanation is far more effective than just explaining how to conduct an operation step-by-step. This is particularly true for transferring knowledge to new problems. For science teaching, this means that explaining how to solve a specific task (e.g. how to calculate the force acting on an object) must include the underlying principle at work (i.e. Newton's laws). Explanation should not finish by elucidating every single step required to arrive at a solution, since learners will have difficulties transferring this knowledge to other problems.

Factor 9: possible tools for increasing adaptation to the explainee's prior knowledge

As discussed above, a common problem of instructional explanations is that learners often just superficially deal with the knowledge that has been explained (Berthold & Renkl, 2010). However, the primary goal of creating effective instructional explanations is reaching a deeper processing of information. There has been considerable research seeking to identify features that help increase the quality of explanation. While all the points discussed above are merely different strategies of explaining, the features highlighted in the following are tools to implement them, particularly adaptation. Adaptation to learners' prior knowledge has been identified as crucial for instructional explanations. Some studies have aimed to operationalise this abstract goal.

Sevian and Gonsalves (2008) created a rubric that essentially includes tools for increasing the quality of explanation. They developed these features based on analysis of explanations from 32 graduate students. Their rubric describes high-inferential overarching categories that are open to interpretation, such as 'connecting to prior knowledge', and connects them to sub-categories such as 'judging the audience's understanding'.

Geelan (2013) accorded with the framework of Wittwer and Renkl (2008) and analysed 16 videos of high school physics teachers' explanations. He chose to evaluate the explanations of exemplary teachers, whose students were particularly successful in external examinations. Geelan (2013) describes six features that most of these model teachers' explanations share:

(1) Shifting from a qualitative start to a mathematical description
(2) Highlighting relevant parts for success in exams
(3) Using analogies
(4) A tendency to use storytelling, especially concerning the history of science
(5) Integrating new technology into their explanations
(6) Humour (as a way to adapt to the students' interests)

Since these characteristics originate from the analysis of outstanding teachers, it might be justified to think of them as being helpful in explaining quality.

Kulgemeyer and Schecker (2013) present another framework that has been used to analyse both the quality of verbal explanations and explanation videos. They describe 45 features that increase the quality of instruction (Kulgemeyer & Peters, 2016; Kulgemeyer & Tomczyszyn, 2015), with 12 of them being identified as core categories. These core categories should be thought of as good indicators for the occurrence of all 45 features.

According to Kulgemeyer and Tomczyszyn (2015), the quality of explanation is particularly increased by:

(1) Presenting numbers as illustrations (instead of leaving formulas unexplained)
(2) Explaining physics concepts in everyday language
(3) Connecting non-verbal elements such as diagrams and pictures in a meaningful manner by showing similarities
(4) Using items such as hands-on experiments
(5) Connecting these items with the explained topic by using analogies (thereby increasing coherence)
(6) Using small demonstrations that are connected to the explained principle
(7) Answering the direct questions of an explainee
(8) Giving reviews by highlighting the relevant parts of the already explained topic
(9) Presenting a summary of relevant parts
(10) Encouraging the explainee, notably by applauding correct answers and stressing that some parts are difficult
(11) Assessing understanding by giving tasks or asking questions other than 'Did you understand that?'
(12) Requesting action from explainees by asking questions or giving prompts

Kulgemeyer and Schecker (2009) describe a 'model of dialogic explaining' that includes these core categories as special applications of greater ideas. This model focuses on the tools that allow adaptation during instructional explanations in science teaching.

According to this model, adaptation requires two components: adaptation to the explainee and adaptation to the subject matter. It is not merely the subject matter that needs to be adapted to the addressee's prerequisites. The subject matter in the explanation differs from the pure scientific subject matter in many ways and cannot be transferred directly to a science classroom. This idea is more of an educational conceptualisation of explanation than a psychological one and comes close to the essence of the model of educational reconstruction (e.g. Duit, Gropengießer, Kattman, Komorek, & Parchmann, 2012).

In addition to these perspectives, the model also provides tools for adaptation. It proposes four 'variables' that must be adjusted to the explainee and the subject matter in each explanation. Adapting the *language level* (ranging from everyday to scientific language) to the explainee must be prioritised, as well as the level of *mathematisation* (ranging from, for example, qualitative descriptions to formulas) and the use of appropriate *representation forms and demonstrations* (ranging from, for example, photos to animations, diagrams, and beyond to items and even experimental setups). The fourth variable is the use of *examples and analogies*, which should be adapted to the explainee, as well. Examples and analogies play a crucial role because they help connect the explained subject matter to learners' prior knowledge (c.f. the idea of 'bridging analogies' (Clement, 1993)). This model has been researched in several empirical studies regarding its validity, e.g.:

- The model has been shown to qualitatively describe how students as explainers adapt their explanations to younger peers (Kulgemeyer & Schecker, 2013).
- In a subsequent study, the model is also found to describe how teachers adapt their instructional explanations after signs of misunderstanding (Kulgemeyer & Tomczyszyn, 2015).
- The model predicts which of two different explanations expert explainers will rate as the better one (Kulgemeyer & Tomczyszyn, 2015).

Kulgemeyer and Riese (2018) use this model to research the relationship between teachers' professional knowledge and their instructional quality in situations involving explaining. They report that content knowledge, mediated by pedagogical content knowledge, supports the quality of teachers' instructional explanations. Additionally, a transmissive view on teaching and learning *decreases* the quality of the instructional explanations.

In summary, we would argue that the features in the lists provided by Geelan (2013), Sevian and Gonsalves (2008), and Kulgemeyer and Tomczyszyn (2015) are essentially expressions of the first five core ideas regarding what makes explaining successful, which we presented at the beginning of the article. Successful explaining is always an interaction between the explainer and explainee, i.e. the basic constructivist idea expressed by the model of Kulgemeyer and Schecker (2009). Most of the features describe the product (*explanation*), and not the process (*interaction*). Research, however, increasingly concludes that the process is what really matters.

Table 1 links the features of the lists provided by Geelan (2013), Sevian and Gonsalves (2008), and Kulgemeyer and Tomczyszyn (2015) to the first five core ideas of explanation (see the section above entitled 'A framework for effective instructional explanations in science teaching'). It is interesting to see that even though none of these authors explicitly mention the five core ideas, all of their features can be assigned to them. We could interpret this as support for the validity of their results because a broad range of studies on instructional explanations support the features they used for analysis.

There is one important exception that requires more discussion. Geelan (2013) and Sevian and Gonsalves (2008) use a feature they call 'storytelling' or 'telling stories', respectively. Ogborn, Kress, Martins, and McGillicuddy (1996) – in one of the first empirical studies on effective instructional explanation in science – describe storytelling as a way of appropriately explaining. Norris, Guilbert, Smith, Hakimelahi, and Phillips (2005) even present a framework for narrative explanations. Based on the studies discussed above, we must be careful about this. On one hand, providing a scientific explanation by telling a story could lead to a better adaptation to the explainee's interests. On the other hand, storytelling also requires the use of information beyond the subject matter. This increases the cognitive load and is a violation of the core principle that explanations should be minimal. Which effect is more important is an open question for further research.

Connections to the idea of basic dimensions in the research on instructional quality

Instructional explanations are one element of science instruction, along with, for example, laboratory work, cooperative learning, and many more components. Laboratory

Table 1. Connecting the features for increasing explanation quality mentioned in three different studies to core ideas of effective explanation.

'Core ideas' of effective instructional explanations	*Sevian & Gonsalves' (2008) (Activities)*	Geelan (2013)	Kulgemeyer and Tomczyszyn (2015) *(Core categories)*
1. Adaptation to content knowledge, misconceptions, interests	• Asking questions • Judging the audience's understanding • Identifying differences between how things appear and how they are • Telling stories	• Storytelling • Humour	• Answering the direct questions of an explainee • Encouraging the explainee, especially by applauding correct answers and stressing that some parts are difficult • Assessing understanding by giving tasks or asking questions other than 'Did you understand that?' • Requesting action from the explainee by asking questions or giving prompts
2. Tools for adaptation	• Use of pictures to illustrate concepts • Use of demonstrations • Mental images • Use of examples • Choice of words • Use of media	• Using analogies • Integrating new technology into their explanations	• Using items such as hands-on experiments • Using small demonstrations that are connected to the explained principle • Explaining physics concepts in everyday language • Presenting numbers as illustrations
3. Highlighting the relevancy	• Placing the research in the context of the larger picture	• Highlighting relevant parts for success in exams	• Giving reviews by noting the relevant parts of the already explained topic • Presenting a summary of relevant parts
4. Structure	• Explicit structure	• Qualitative start, quantitative ending	
5. Minimal explanations			• Connecting non-verbal elements such as diagrams and pictures in a meaningful manner by showing similarities • Connecting items with the explained topic by showing an analogy (thereby increasing coherence)

coursework and demonstrations are prominent teaching tools, and – compared to instructional explanations – considerable research exists regarding how to foster their effectiveness (Abrahams & Millar, 2008; Hofstein & Kind, 2012; Hofstein & Lunetta, 2004). Surprisingly, the research on effective instructional explanations has never been connected to the general research on instructional quality. From our perspective, different terms have been used for what are sometimes very similar ideas. In this section, we want to discuss possible links based on the core ideas of instructional explanations presented above and a very influential concept regarding general instructional quality: the idea of basic dimensions.

Since the 1960s (e.g. Carroll, 1963), there has been considerable research concerning the question of whether general features of successful teaching can be identified. Many studies have had the goal of identifying links between structures in teaching and student outcomes ('process/product research' (Brophy, 2000)). Some meta-analyses have been conducted (e.g. Hattie, 2009; O'Neill, 1988; Seidel & Shavelson, 2007). Over the years, three basic dimensions of instructional quality have emerged that have recently been used in many studies. Following the introduction of the 'Classroom Learning Assessment Scoring System' (Pianta, Hamre, & Mintz, 2012), they are sometimes

referred to as 'emotional support', 'classroom organisation', and 'instructional support'. In European research, they are more often known as (1) classroom management, (2) constructive support, and (3) cognitive activation (Klieme, Schümer, & Knoll, 2001). However, these meanings differ just slightly (Dorfner, Förtsch, & Neuhaus, 2017). There are many very influential studies that have been conducted recently that rely on the three basic dimensions. These include three major projects: 'The Competence of Teachers, Cognitively Activating Instruction and Development of Students' Mathematical Literacy' study (COACTIV) (Baumert et al., 2010) in mathematics, the 'Professional Knowledge in Science' (ProwiN) project (Cauet, Liepertz, Kirschner, Borowski, & Fischer, 2015) in science and the 'Quality of Instruction in Physics' (QuIP) project (Fischer, Neumann, Labudde, & Viiri, 2014). The basic dimensions include the following ideas:

- *Classroom management* refers to strategies that lead to a well-organised classroom. Examples are a high amount of time spent on task or the existence of explicit rules. Of course, these characteristics seem important for situations involving explanation, as well; however, they are not specific for instructional explanations, but rather general.
- *Constructive support* includes features such as a patient teacher, a positive environment that accepts mistakes as learning opportunities, and appropriate feedback. It often also refers to more cognitive aspects (Einsiedler & Hardy, 2010). Scaffolding is an element of constructive support and involves transforming subject matter into a content structure for instruction (which, for example, includes determining basic concepts). We see some parallels to the research on effective instructional explanations here.
- *Cognitive activation* is likely the most researched of the three basic dimensions. The three major projects – COACTIV, ProwiN, and QuIP – focus on cognitive activation first. This includes challenging learning tasks, cognitive conflicts, considering prior knowledge, and motivating students to participate in, for example, questioning techniques. There seem to be possible connections to the research on effective instructional explanations.

The three basic dimensions are considered to be independent of the teaching situation or content. From the three projects (COACTIV, ProwiN, and QuIP), there is substantial evidence that cognitive activation has a major impact on students' achievements. Considering this, there should be connections to the research on effective instructional explanations. After all, these explanations are just one element of instruction in general.

Indeed, some connections of the core ideas of effective instructional explanation to the three basic dimensions can be identified. *Adaptation to content knowledge, misconceptions, and interests* (Core Idea 1) are at the centre of the general idea behind cognitive activation. Cognitive activation essentially means that students follow instruction and engage in deep-processing while doing so. Adaptation to content knowledge and possible misconceptions is the basic prerequisite for deeper processing, while adaptation to interests provides better chances of students being motivated. The *tools for adaptation* (Core Idea 2), therefore, are very detailed descriptions on how to achieve

cognitive activation during instructional explanations. We also know from the research discussed above that *highlighting the relevancy* (Core Idea 3) is effective because it avoids the impression of redundancies and uselessness and, thus, leads to deeper processing of information. This core idea also fits very well with the basic dimension of cognitive activation.

Structure (Core Idea 4) fits well with the basic dimension of constructive support. Through the cognitive element of this basic dimension, providing structure in explanation enables students to better build structures through their mental imagery, as well. Helping students build mental models is a major part of constructive support. That is why *minimal explanations* (Core Idea 5) can also be linked to this basic dimension. Minimal explanations include a high coherence and a low cognitive load (which, specifically, means avoiding redundancies). Research on minimal explanations implies that this is helpful for constructing mental models.

It may not be surprising that the core ideas of effective instructional explanations fit so well with the more general basic dimensions of instructional quality. However, it is somewhat surprising that research in each field currently does not refer to the other. Currently, we can differentiate between three levels of research on effective instructional explanations. The first level is the applied research that aims at finding features of successful explanations, resulting in lists of features like those of Geelan (2013) and Kulgemeyer and Tomczyszyn (2015), or rubrics like the one of Sevian and Gonsalves (2008). This kind of research focuses mostly on the explanation itself and has just minor references to underlying psychological theories. The second level is the research that is represented by the work of Wittwer and Renkl (2008) and the studies we discussed in the first chapter. All these studies refer to psychological theories, such as the cognitive load theory. They do not aim at finding concrete features, but rather at testing the underlying theories and describing high-inferential characteristics of exemplary explanation. The third level of research is well aware that all the features and core ideas discussed are merely expressions of the basic dimensions of instructional quality in one specific teaching situation. We believe that this level provides considerable impact for studies on instructional explanations in science education. This would also be true for other teaching situations in science. As mentioned, practical coursework and demonstrations are teaching tools in which significant research has been done regarding different ways to increase their impact. The research on practical coursework and demonstrations might also find connections to the basic dimensions of instructional quality.

Implications for further research

In this article, we propose a framework for how instructional explanations should be given in science teaching based on the studies presented above. To discuss the framework, we review the literature on instructional explanations with a focus on science education. However, we are aware that the evidence for this framework differs in its quality. Furthermore, we want to highlight the fact that following this framework does not automatically lead to better understanding. That would be adopting the transmissive view regarding explanation, which has been proven wrong several times. Rather, following the framework increases the likelihood that an individual

constructs meaning from an explanation. Additionally, the framework gives advice based on the current state of research, and further studies might add very important aspects.

One of the major problems we see in the research on instructional explanations is that most of the basic results come from studies in educational psychology. That leads to a selection bias and important restrictions of the analysis:

(1) Most of the samples in the analysed studies consist of psychology students, and we cannot be sure that high school students in science courses behave the same way.
(2) Most of the explained topics are not related to school science subjects. In fact, there is a surprisingly high number of topics from computer science. We cannot be sure if there is a domain-specific way of explaining science beyond the state of research. The role of misconceptions concerning the adaptation of instructional explanations has hardly been researched in psychology. That would be a major challenge for science education research. As a result, we identify those studies in the references that clearly deal with scientific topics with an asterisk.
(3) In our opinion, the studies from educational psychology are applicable to science teaching in general. However, there are some science-specific problems that require further research, e.g. (a) how to deal with misconceptions in instructional explanations, (b) which principles in science are sophisticated enough to necessitate an instructional explanation, or (c) which principles allow self-explanations. Generally, we do not know enough about the 'culture of explaining' in science teaching. A culture of explaining means how to implement instructional explanations in a teaching sequence, how to construct them, and when to provide them. It also includes the questions of which scientific principles should be introduced by an instructional explanation and in which topics in science might different teaching techniques be more promising.
(4) Most of the sample sizes are rather small, comparing two groups of about 50 students each. That helps researchers find large or medium effects, but can large or even medium effects be expected after a short explanation?
(5) Science explanation videos are an emerging topic in science teaching. They are provided online for free by various sources and deal with all major science topics (Kulgemeyer & Peters, 2016). Can the criteria for instructional explanations be applied to these videos (Kulgemeyer, 2018)? How can they be integrated into science teaching and how can teachers deal with the problem that this medium does not allow an adaptation of the explanation approach?
(6) How does the medium affect the core ideas of what makes instructional explanations effective in general? Some of the results come from studies dealing with verbal explanations, and some come from studies dealing with texts. Textbooks offer explanations, and it is unclear if they require different core ideas.
(7) Does the validity or the scientific status of an instructional explanation influence how students perceive it? Does the type of scientific explanation used affect a learner's perception, and does that influence the effectiveness of an instructional explanation?

We consider the model to be a promising framework for giving instructional explanations both as a teacher and as a science communicator. It also is a promising model for research. Of course, we do not know enough about how the core ideas of instructional explanation interact with one another. Research usually focuses on variable control, and therefore on just one of the core ideas presented. It would be a challenge for future research to design teaching sequences based on the model and research it as a whole.

Instructional explanations are one teaching method out of many, and we can only support Geelan's (2012) view that many more studies on this very common practice must be conducted. For example, we know too little about their success criteria. In literature, explanation is often discussed from a very critical perspective ('drill and kill', or 'didactical teaching'), and the constructivist view on instruction is often not mentioned. However, just like Wittwer and Renkl (2008), we agree with Acuña, Rodicio, & Sánchez's (2011) statement that 'In fact, when adjusted to the level of learners' knowledge and provided after learners' misunderstandings have been pointed out, instructional explanations are effective indeed' (Acuña et al., 2011, p. 438).

Disclosure statement

No potential conflict of interest was reported by the author.

Notes on contributor

Christoph Kulgemeyer is an associate professor ('Privatdozent') at the Institute of Science Education, Physics Education Group, University of Bremen. He conducts research on instructional quality in physics, performance-based assessment methods, and the connection between teacher education and teaching performance. Over the last six years, he wrote his 'habilitation' thesis on instructional explanations in science teaching.

ORCID

Christoph Kulgemeyer http://orcid.org/0000-0001-6659-8170

References

*Abrahams, I., & Millar, R. (2008). Does practical work really work? A study of the effectiveness of practical work as a teaching and learning method in school science. *International Journal of Science Education*, *30*(14), 1945–1969.

Acuña, S., García-Rodicio, H., & Sánchez, E. (2011). Fostering active processing of instructional explanations of learners with high and low prior knowledge. *European Journal of Psychology of Education*, *26*, 435–452.

Altmann, A., & Nückles, M. (2017). Empirische Studie zu Qualitätsindikatoren für den diagnostischen Prozess [Empirical studies on quality criteria for a diagnostic process]. In A. Südkamp & A.-K. Praetorius (Eds.), *Diagnostische Kompetenz von Lehrkräften: Theoretische und methodische Weiterentwicklungen* (pp. 134–141). Münster: Waxmann.

Alty, J.L., & Coombs, M.J. (1981). Communicating with university computer users: A case study. In M.J. Coombs & J.L. Alty (Eds.), *Computing skills and the user interface* (pp. 7–71). London, UK: Academic Press.

Anderson, J.R., Corbett, A.T., Koedinger, K.R., & Pelletier, R. (1995). Cognitive tutors: Lessons learned. *The Journal of the Learning Sciences*, *4*, 67–207.

Baumert, J., Kunter, M., Blum, W., Brunner, M., Voss, T., Jordan, A., … Tsai, Y.M. (2010). Teachers' mathematical knowledge, cognitive activation in the classroom, and student progress. *American Educational Research Journal*, *47*(1), 133–180.

*Belova, N., Rundgren, S., & Eilks, I. (2015). Advertising and science education: A multi-perspective review of the literature. *Studies in Science Education*, *51*(2), 169–200.

*Berger, R., & Hänze, M. (2015). Impact of expert teaching quality on novice academic performance in the jigsaw cooperative learning method. *International Journal of Science Education. 37*(2), 294–320.

Berghmans, I., Michiels, L., Salmon, S., Dochy, F., & Struyven, K. (2014). Directive versus facilitative peer tutoring? A view on students' appraisal, reported learning gains and experiences within two differently-tutored learning environments. *Learning Environments Research*,*17*, 437–459.

*Berland, L.K., & Reiser, B.J. (2008). Making sense of argumentation and explanation. *Science Education 93*, 26–55.

*Berland, L.K., & McNeill, K.L. (2012). For whom is argument and explanation a necessary distinction? A response to Osborne and Patterson. *Science Education*, *96*(5), 808–813.

Berthold, K. (2012). Explanatory support for learning. In N. Seel (Ed.), *Encyclopedia of the sciences of learning* (pp. 1241–1244). New York, NY: Springer.

Berthold, K., & Renkl, A. (2010). How to foster active processing of explanations in instructional communication. *Educational Psychology Review*, *22*, 25–40.

Berthold, K., Röder, H., Knörzer, D., Kessler, W., & Renkl, A. (2011). The double-edged effects of explanation prompts. *Computers in Human Behavior*, *27*, 69–75.

Black, J.B., Carroll, J.M., & McGuigan, S.M. (1987). What kind of minimal instruction is the best? In J. M. Carroll & P.P. Tanner (Eds.), *Proceedings of CHI+Gi´87* (pp. 159–162). New York, *human factors in computing systems and graphics interface*NY: ACM.

*Braaten, M., & Windschitl, M. (2011). Working toward a stronger conceptualization of scientific explanation for science education. *Science Education 95*(4), 639–669.

*Brigandt, I. (2016). Why the difference between explanation and argument matters to science education. *Science & Education*, *25*(3), 251–275.

Brophy, J. (2000). *Teaching*. Brussels, Belgium: UNESCO International Academy of Education (IAE).

*Cabello, V. (2013). *Developing skills to explain scientific concepts during initial teacher education: The role of peer assessment*. Dundee, UK: University of Dundee.

Carroll, J. (1963). A model of school learning. *Teachers College Record*, *64*(8), 723–733.

Carroll, J. (1990). *The Nurnberg Funnel. Designing minimalist instruction for practical computer skill*. Cambridge: MIT Press.

*Champagne, A.B., Klopfer, L.E., & Gunstone, R.F. (1982). Cognitive research and the design of science instruction. *Educational Psychologist*, *17*, 31–53.

*Cauet, E., Liepertz, S., Kirschner, S., Borowski, A., & Fischer, H.E. (2015). Does it matter what we measure? Domain-specific professional knowledge of physics teachers. *Revue Suisse des sciences de l'éducation*, *37*(3), 462–479.

Charalambous, C.Y., Hill, H.C., & Ball, D.L. (2011). Prospective teachers' learning to provide instructional explanations: How does it look and what might it take? *Journal of Mathematics Teacher Education*, *14*(6), 441–463.

Chi, M.T.H. (1996). Constructing self-explanations and scaffolded explanations in tutoring. *Applied Cognitive Psychology*, *10*, 33–49.

Chi, M.T.H. (2000). Self-explaining expository texts: The dual processes of generating inferences and repairing mental models. In R. Glaser (Ed.), *Advances in instructional psychology* (pp. 161–238). Mahwah, NJ: Lawrence Erlbaum Associates.

Chi, M.T.H. (2009). Active-constructive-interactive: A conceptual framework for differentiating learning activities. *Topics in Cognitive Science*, *1*(1), 73–105.

Chi, M.T.H., Bassok, M., Lewis, M.W., Reimann, P., & Glaser, R. (1989). Self-explanations: How students study and use examples in learning to solve problems. *Cognitive Science*, *13*, 145–182.

Chi, M.T.H., de Leeuw, N., Chiu, M.H., & LaVancher, C. (1994). Eliciting self-explanations improves understanding. *Cognitive Science*, *18*, 439–477.

Chi, M.T.H., Siler, S.A., & Jeong, H. (2004). Can tutors monitor students understanding accurately? *Cognition and Instruction*, *22*, 363–387.

Chi, M.T.H., Siler, S.A., Jeong, H., Yamauchi, T., & Hausmann, R.G. (2001). Learning from human tutoring. *Cognitive Science*, *25*, 471–533.

*Clement, J. (1993). Using bridging analogies and anchoring instructions to deal with students' Preconceptions in physics. *Journal of Research in Science Teaching*, *30*(10), 1241–1257.

Connell, L., & Keane, M.T. (2004). What plausibility affects plausibility? Concept-coherence and distributional word coherence as factors influencing plausibility judgments. *Memory and Cognition*, *32*, 185–197.

*Dagher, Z., & Cossman, G. (1992). Verbal explanations given by science teachers: Their nature and implications. *Journal of Research in Science Teaching*, *29*, 361–374.

*Diakidoy, I.N., Kendeou, P., & Ioannides, C. (2003). Reading about energy: The effects of text structure in science learning and conceptual change. *Contemporary Educational Psychology*, *28*, 335–356.

Dorfner, T., Förtsch, C., & Neuhaus, B. (2017). Die methodische und inhaltliche Ausrichtung quantitativer Videostudien zur Unterrichtsqualität im mathematisch-naturwissenschaftlichen Unterricht [The Methodical and Content-related Orientation of Quantitative Video Studies on Instructional Quality in Mathematics and Science Education]. *Zeitschrift für Didaktik der Naturwissenschaften*. doi: 10.1007/s40573-017-0058-3.

Duffy, G., Roehler, L., Meloth, M., & Vavrus, L. (1986). Conceptualizing instructional explanation. *Teaching and Teacher Education*, *2*, 197–214.

*Duit, R., & Treagust, D.F. (2003). Conceptual change: A powerful framework for improving science teaching and learning. *International Journal of Science Education*, *25*(6), 671–688.

*Duit, R., & Pfundt, H. (2009). *Bibliography - STCSE students' and teachers' Conceptions and science education*. Kiel, Germany: IPN

*Duit, R., Gropengießer, H., Kattman, U., Komorek, M., & Parchmann, I. (2012). The model of educational reconstruction – A framework for omproving teaching and learning science. In D. Jorde & J. Dillon (Eds.), *Science education research and practice in europe: retrospective and prospective* (pp. 13–37). Dordrecht, The Netherlands: Springer.

Dutke, S., & Reimer, T. (2000). Evaluation of two types of online help for application software. *Journal of Computer-Assisted Learning*, *16*, 307–315.

Einsiedler, W., & Hardy, I. (2010). Kognitive Strukturierung im Unterricht: Einführung und Begriffsklärungen [Cognitive Structuring in Teaching: Instrudocution and Clarification of the concept]. *Unterrichtswissenschaft*, *38*. 94–209.

*Fechner (2009). *Effects of context-oriented learning on student interest and achievement in chemistry education*. Berlin, Germany: Logos.

*Ford, D. (2009). Promises and challenges for the use of adapted primary literature in science curricula: Commentary. *Research in Science Education 39*. 385–390

*Fischer, H., Neumann, K., Labudde, P., & Viiri, J. (2014). (Eds.), *Quality of instruction in physics. Comparing Finland, Germany and Switzerland*. Münster, Germany: Waxmann.

Gage, N. (1968). The microcriterion of effectiveness in explaining. In N. Gage (Ed.), *Explorations of the teacher's effectiveness in explaining*, Technical *Report No. 4*, (pp.1–8). Stanford, CA: Stanford Center for Research and Developement in Teaching.

García-Sanchez, J.-N., & Fidalgo-Redondo, R. (2006). Effects of two types of self-regulatory instruction programs on students with learning disabilities in writing products, processes, and self-efficacy. *Learning Disability Quarterly 29*, 181–211.

*Hänze, M., & Berger, R. (2007). Cooperative learning, motivational effects and student characteristics: An experimental study comparing cooperative learning and direct instruction in 12th grade physics classes. *Learning and Instruction 17*, 29–41.

*Geelan, D. (2012). Teacher explanations. In B. Fraser, K. Tobin, & C. McRobbie (Eds.), *Second international handbook of science education* (pp. 987–999). Dordrecht, The Netherlands: Springer.

*Geelan, D. (2013). Teacher explanation of physics concepts: A video study. *Research in Science Education 43*, 1751–1762.

Hattie, J. (2009). *Visible Learning*. London, UK: Routledge.

Hempel, C., & Oppenheim, P. (1948). Studies in the logic of explanation. *Philosophy of Science*, *15*(2), 135–175.

Herppich, S., Wittwer, J., Nückles, M., & Renkl, A. (2014). Addressing knowledge deficits in tutoring and the role of teaching experience: Benefits for learning and summative assessment. *Journal of Educational Psychology*, *106*(4), 934–945.

*Hiebert, J., Gallimore, R., Garnier, H., Givvin, K.B., Hollingsworth, H., & Jacobs, J. (2003). *Teaching mathematics in seven countries: Results from the TIMSS 1999 video study*. Washington: U.S. Department of Education, National Center for Education Statistics.

Hmelo-Silver, C.E. (2004). Problem-based learning: What and how do students learn? *Educational Psychology Review*, *16*, 235–266.

*Hofstein, A., & Lunetta, V. (2004). The laboratory in science education: Foundations for the twenty-first century. *Science Education*, *88*(1), 28–54.

*Hofstein, A., & Kind, P.M. (2012). Learning in and from science laboratories. In B. Fraser, K. Tobin, & C. McRobbie (Hrsg.): *Second international handbook of science education* (pp.189–207). Dordrecht, The Netherlands: Springer.

Kalyuga, S. (2007). Expertise reversal effect and its implications for learner-tailored instruction. Educational *Psychology Review*, *19*, 509–539.

Kalyuga, S., Ayres, P., Chandler, P., & Sweller, J. (2003). The expertise reversal effect. *Educational Psychologist*, *38*, 23–31.

Kalyuga, S., & Renkl, A. (2010). Expertise reversal effect and its instructional implications: Introduction to the special issue. *Instructional Science*, *38*, 209–215.

Kamalski, J., Sanders, T., & Lentz, L. (2008). Coherence marking, prior knowledge and comprehension of informative and persuasive text: Sorting things out. *Discourse Processes*, *45*, 323–345.

*Kendeou, P., & van Den Broek, P. (2007). The effects of prior knowledge and text structure on comprehension processes during reading of scientific texts. *Memory & Cognition*, *35*, 1567–1577.

King, A. (1999). Discourse patterns for mediating peer learning. In A. O'Donnell & A. King (Eds.), *Cognitive perspectives on peer learning* (pp. 3–37). Mahwah, NJ: Erlbaum.

Kitcher, P. (1981). Explanatory Unification. *Philosophy of Science*, *48*(4), 507–531.

Klieme, E., Schümer, G., & Knoll, S. (2001). Mathematikunterricht in der Sekundarstufe I: „Aufgabenkultur' und Unterrichtsgestaltung [Mathematic instruction in lower secondary: 'culture of learning tasks' and quality of instruction]. In Bundesministerium für Bildung und Forschung (BMBF) (Hrsg.), *TIMSS – Impulse für Schule und Unterricht. Forschungsbefunde, Reforminitiativen, Praxisberichte und Video-Dokumente* (pp. 43–57). Bonn: BMBF.

Korthagen, F.A.J., & Kessels, J.P.A.M. (1999). Linking theory and practice: Changing the pedagogy of teacher education. *Educational Researcher*, *28*(4), 4–17.

*Kulgemeyer, C., & Schecker, H. (2009). Kommunikationskompetenz in der Physik: Zur Entwicklung eines domänenspezifischen Kompetenzbegriffs. *Zeitschrift für Didaktik der Naturwissenschaften 15*, 131–153

*Kulgemeyer, C., & Schecker, H. (2013). Students explaining science – Assessment of science communication competence. *Research in Science Education*, *43*, 2235–2256.

*Kulgemeyer, C., & Tomczyszyn, E. (2015). Physik erklären – Messung der Erklärensfähigkeit angehender Physiklehrkräfte in einer simulierten Unterrichtssituation [Explaining physics –

measuring teacher trainees' Explaining skills using a simulated teaching setting]. *Zeitschrift für Didaktik der Naturwissenschaften*, *21*(1), 111–126.

*Kulgemeyer, C., & Peters, C. (2016). Exploring the explaining quality of physics online explanatory videos. *European Journal of Physics*, *37*(6), 1–14.

*Kulgemeyer, C. (2018). A Framework of effective science explanation videos informed by criteria for instructional explanations. *Research in Science Education*, 1–22. DOI: 10.1007/s11165-018-9787-7

*Kulgemeyer, C., & Riese, J. (2018). From professional knowledge to professional performance: The impact of CK and PCK on teaching quality in explaining situations. *Journal of Research in Science Teaching*, 1–26. DOI: 10.1002/tea.21457.

Lehman, S., & Schraw, G. (2002). Effects of coherence and relevance on shallow and deep text processing. *Journal of Educational Psychology*, *94*, 738–750.

Leinhardt, G., & Steele, M.D. (2005). Seeing the complexity of standing to the side: Instructional dialogues. *Cognition and Instruction*, *23*, 87–163.

Martin, J. (1970). *Explaining, understanding, and teaching*. New York, NY: McGraw-Hill.

Mayer, R.E., & Moreno, R. (2003). Nine ways to reduce cognitive load in multimedia learning. *Educational Psychologist*, *38*, 43–52.

Mayer, R.E., Moreno, R., Boire, M., & Vagge, S. (1999), Maximizing constructivist learning from multimedia communications by minimizing cognitive load, *Journal of Educational Psychology*, *91*, 638–643.

McNamara, D.S., & Kintsch, W. (1996). Learning from texts: Effects of prior knowledge and text coherence. *Discourse Processes*, *22*, 247–288.

*Merzyn, G. (2005). Junge Lehrer im Referendariat [Young teachers in teacher training]. *Der mathematische und naturwissenschaftliche Unterricht*, *58*(1), 4–7.

*McNeill, K. (2009). Teachers' use of curriculum to support students in writing scientific arguments to explain phenomena. *Science Education*, *93*(2), 233–268.

Nathan, M., & Koedinger, K. (2000). An investigation of teachers's beliefs of students' algebra development. *Cognition & Instruction*, *40*, 209–237.

Nathan, M., & Petrosino, A. (2003). Expert blind spot among preservice teachers. *American Educational Research Journal*, *40*(4), 905–928.

Nickerson, R.S. (1999). How we know – And sometimes misjudge – What others know: Imputing one's own knowledge to others. *Psychological Bulletin*, *125*, 737–759.

*Norris, S.P., Guilbert, S.M., Smith, M.L., Hakimelahi, S., & Phillips, L.M. (2005). A theoretical framework for narrative explanation in science. *Science Education*, *89*(4), 535–563.

Nückles, M., Wittwer, J., & Renkl, A. (2005). Information about a layer- son's knowledge supports experts in giving effective and efficient online advice to laypersons. *Journal of Experimental Psychology: Applied*, *11*, 219–236.

O'Neill, G.P. (1988). Teaching effectiveness: A review of the research. *Canadian Journal of Education*, *13*(1), 162–185.

*Ogborn, J., Kress, G., Martins, I., & McGillicuddy, K. (1996). *Explaining science in the classroom*. Buckingham, UK: Open University Press.

*Phillips, L., & Norris, S. (2009). Bridging the gap between the language of science and the language of school science through the use of adapted primary literature. *Research in Science Education 39*. 313–319

*Osborne, J.F., & Patterson, A. (2011). Scientific argument and explanation: A necessary distinction? *Science Education*, *95*(4), 627–638.

Pianta, R.C., Hamre, B.K., & Mintz, S. (2012). *Classroom assessment scoring system: Secondary manual*. Charlottesville, VA: Teachstone.

*Posner, G.J., Strike, K.A., Hewson, P.W., & Gertzog, W.A. (1982). Accommodation of a scientific conception: Toward a theory of conceptual change. *Science Education*, *66*, 211–227.

References (Asterisk: Articles from science education research)

Renkl, A. (2002). Worked-out examples: Instructional explanations support learning by self-explanations. *Learning and Instruction*, *12*, 529–556.

Renkl, A., Wittwer, J., Große, C., Hauser, S., Hilbert, T., Nückles, M., & Schworm, S. (2006). Instruktionale Erklärungen beim Erwerb kognitiver Fertigkeiten: Sechs Thesen zu einer oft vergeblichen Bemühung [Instructional explanations and the achievement of cognitive abilities: Six hypotheses on a failing attempt]. In I. Hosenfeld (Ed.): *Schulische Leistung. Grundlagen, Bedingungen, Perspektiven* (205–223). Münster, Germany: Waxmann.

Rescher, N. (1962). The stochastic revolution and the nature of scientific explanation. *Synthese 14*, 200–215.

Roelle, J., Berthold, K., & Renkl, A. (2014). Two instructional aids to optimise processing and learning from instructional explanations. *Instructional Science*, *42*, 207–228.

Ruben, D. (1990). *Explaining Explanation*. London, UK: Routledge.

Sánchez, E., García Rodicio, H., & Acuña, S.R. (2009). Are instructional explanations more effective in the context of an impasse? *Instructional Science*, *37*, 537–563.

*Scott, P., Asoko, H., & Driver, R. (1992): Teaching for conceptual change - A review of strategies. In R. Duit, F. Goldberg, & H. Niedderer (Eds.), *Research in physics learning: Theoretical issues and empirical studies* (pp. 310–329). Kiel: IPN.

Seidel, T., Blomberg, G., & Renkl, A. (2013). Instructional strategies for using video in teacher education. *Teaching and Teacher Education*, *34*, 56–65.

Seidel, T., & Shavelson, R.J. (2007). Teaching effectiveness research in the past decade: The role of theory and research design in disentangling meta-analysis results. *Review of Educational Research*, *77*(4), 454–499

*Sevian, H., & Gonsalves, L. (2008). Analysing how scientists explain their research: A rubric for measuring the effectiveness of scientific explanations. *International Journal of Science Education*, *30*(11), S441–S1467.

Sweller, J., & Chandler, P. (1991). Evidence for cognitive load theory. *Cognition and Instruction*, *8*, 351–362.

*Talanquer, V. (2007). Explanations and teleology in chemistry education. *International Journal of Science Education*, *29*(7), 853–870.

Tomlinson, P.D., & Hunt, D.E. (1971). Differential effects of rule-example order as a function of learner conceptual level. *Canadian Journal of Behavioural Science*, *3*(3), 237–245.

*Treagust, D., & Harrison, A. (1999). The genesis of effective science explanations for the classroom. In J. Loughran (Ed.): *Researching teaching: Methodologies and practices for understanding pedagogy* (pp. 28–43). Abingdon, VA: Routledge.

*Velentzas, A., & Halkia, K. (2018). Scientific explanations in Greek upper secondary physics textbooks. *International Journal of Science Education*, *40*(1), 90–108.

*Trommler, F., Gresch, H., & Hammann, M. (2018). Students' reasons for preferring teleological explanations. *International Journal of Science Education*, *40*(2), 59–187.

Webb, N, & Farivar, S. (1999). Developing productive group interaction in middle school mathematics. In (Eds.), (pp. 117-149).London: Lawrence Erlbaum.

Webb, N.M. (1992). Testing a theoretical model of student interaction and learning in small groups. In R. Hertz-Lazarowitz & N. Miller (Eds.), *Interaction in cooperative groups: The theoretical anatomy of group learning* (pp. 102–119). New York, NY: Cambridge University Press.

Webb, N.M., & Palinscar, A.S. (1996). Group processes in the classroom. In D.C. Berliner & R.C. Calfee (Eds.), *Handbook of educational psychology* (pp. 841–873). New York: Simon & Schuster Macmillan.

Webb, N.M., Ing, M., Kersting, N., & Nemer, K.M. (2006). Help seeking in cooperative learning groups. In S.A. Karabenick & R.S. Newman (Eds.), *Help seeking in academic settings: Goals, groups, and contexts* (pp. 45–88). Mahwah, NJ: Lawrence Erlbaum Associates.

Webb, N.M., Troper, J.D., & Fall, R. (1995). Constructive activity and learning in collaborative small groups. *Journal of Educational Psychology*, *87*, 406–423.

Weiss, I.R., & Parsley, J.D. (2004). What is high-quality instruction? *Educational Leadership*, *65*(1), 24–28.

*Wilson, H., & Mant, J. (2011a). What makes an exemplary teacher of science? The pupils' perspective. *School Science Review*, *93*(342), 121–125.

*Wilson, H., & Mant, J. (2011b). What makes an exemplary teacher of science? The teachers' perspective. *School Science Review*, *93*(342), 115–119.

Wittwer, J., & Ihme, N. (2014). Reading skill moderates the impact of semantic similarity and causal specificity on the coherence of explanations. *Discourse Processes*, *51*, 143–166.

Wittwer, J., Nückles, M., Landmann, N., & Renkl, A. (2010). Can tutors be supported in giving effective explanations? *Journal of Educational Psychology*, *102*, 74–89.

Wittwer, J., & Renkl, A. (2008). Why instructional explanations often do not work: A framework for understanding the effectiveness of instructional explanations. *Educational Psychologist*, *43*(1), 49–64.

*Yip, C. (2009) Causal and teleological explanations in biology. *Journal of Biological Education*, *43* (4), 149–151.

*Yao, J., & Guo, Y. (2017). Validity evidence for a learning progression of scientific explanation. *Journal of Research in Science Teaching*, *55*(2), 299–317.